

TITRE PROFESSIONNEL

CONCEPTEUR / DÉVELOPPEUR

D'APPLICATIONS

By CHEVRIER Clément

2023 - 2024



Présentation



- Clément CHEVRIER
- Alternant chez EvoFlux depuis septembre 2022
- Prestations de services => logiciels de gestion
- 2 salariés + 2 alternant



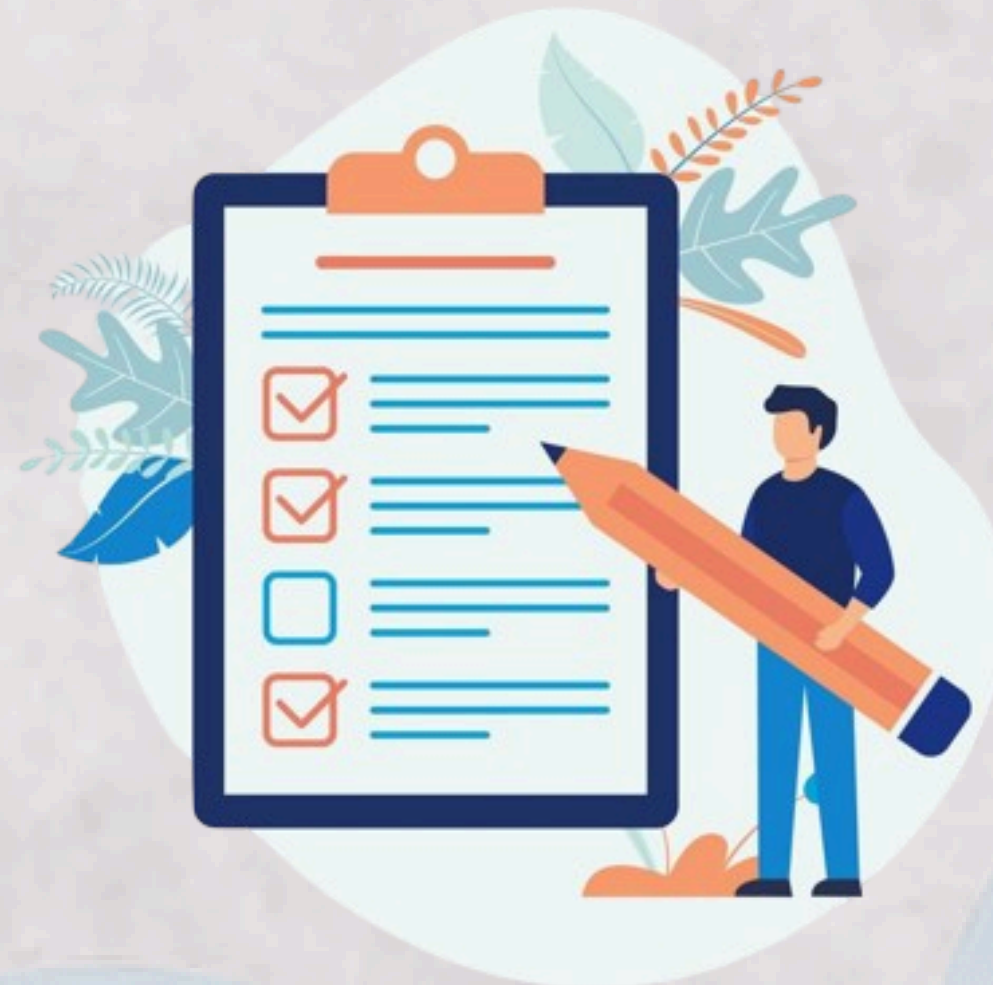
Resume

Interactive questionnaire => Serious Game
Quick diagnosis of management processes



Fonctionnalités

- Questionnaire interactif
- Scénarios et de situations basés sur les réponses
- Fournir un score personnalisé
- Formulaire de contact





Maquettage

Charte Graphique

# 9DCDD1	# 244C7B	# 57A8B0
# 303170	# E2DEDF	# 176585

Police: lisible

<=
Zoning

WireFrame

=>

Questions Préliminaires (Q1-6)

Questions préliminaires (Q1-6)

Texte

Texte

Texte

Texte

Texte

Texte

Bouton

Autres Questions (Unique Q7-19)

Autres Questions (Unique Q7-19)

Texte

Texte

Bouton

Bouton

Bouton

Bouton

Bouton

Questions Préliminaires (Q1-6)

Questions préliminaires (Q1-6)

Titre

Question 1 :

- Réponse 1
- Réponse 2
- Réponse 3
- Réponse 4

Question 2 :

- Réponse 1
- Réponse 2
- Réponse 3
- Réponse 4

Question 3 :

- Réponse 1
- Réponse 2
- Réponse 3
- Réponse 4

Question 4 :

- Réponse 1
- Réponse 2
- Réponse 3
- Réponse 4

Question 5 :

- Réponse 1
- Réponse 2
- Réponse 3
- Réponse 4

Question 6 :

- Réponse 1
- Réponse 2
- Réponse 3
- Réponse 4

Soumettre

Autres Questions (Unique Q7-19)

Autres Questions (Unique Q7-19)

Titre

Titre

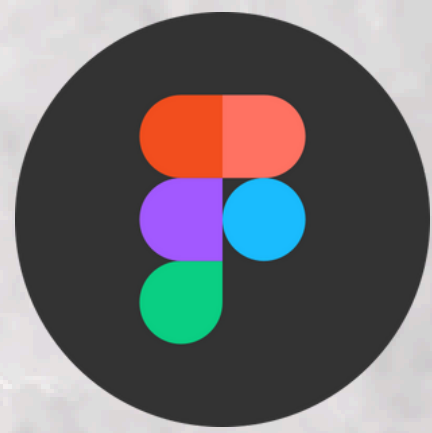
Réponse 1

Réponse 2

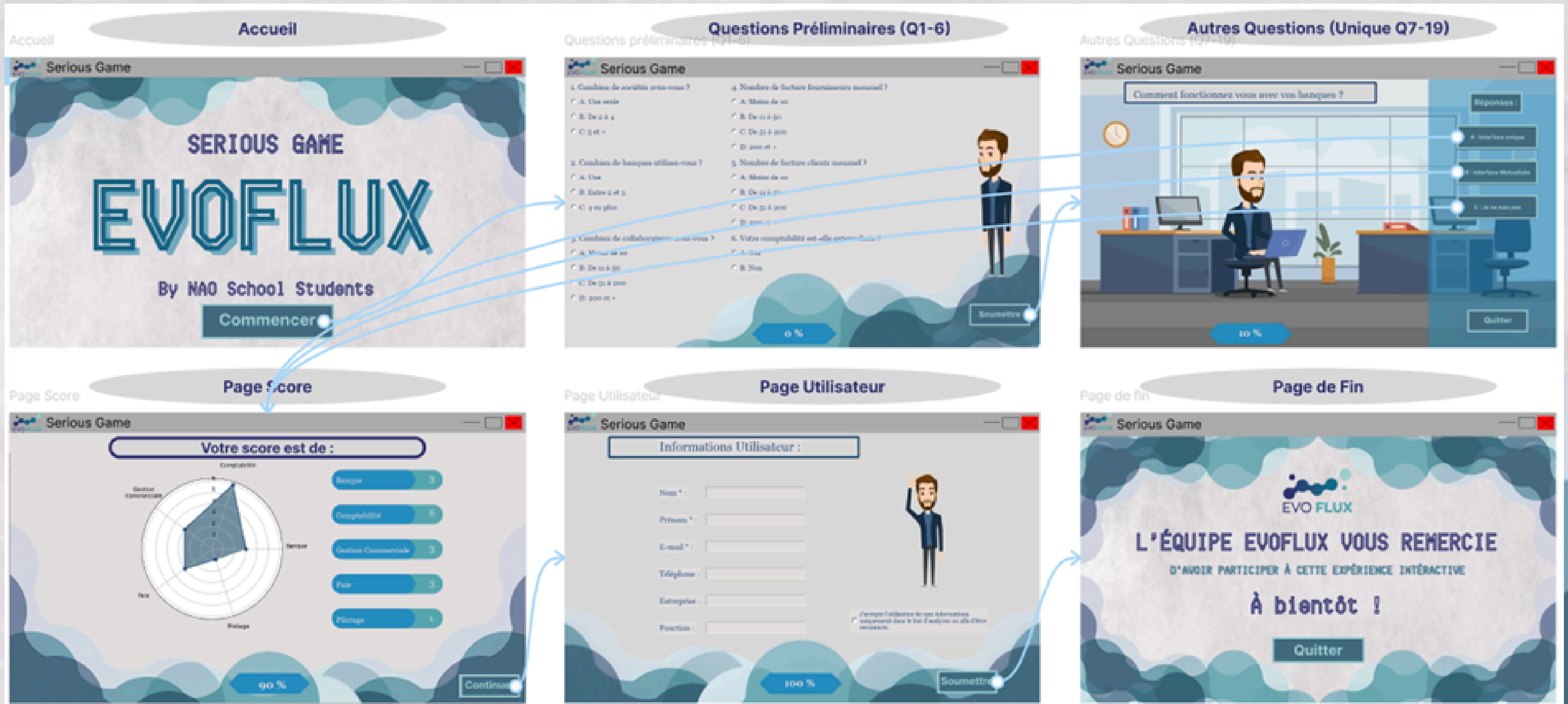
Réponse 3

Réponse 4

Quitter

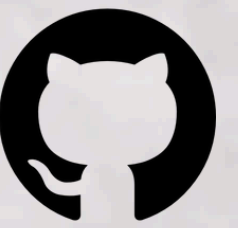


Maquettage



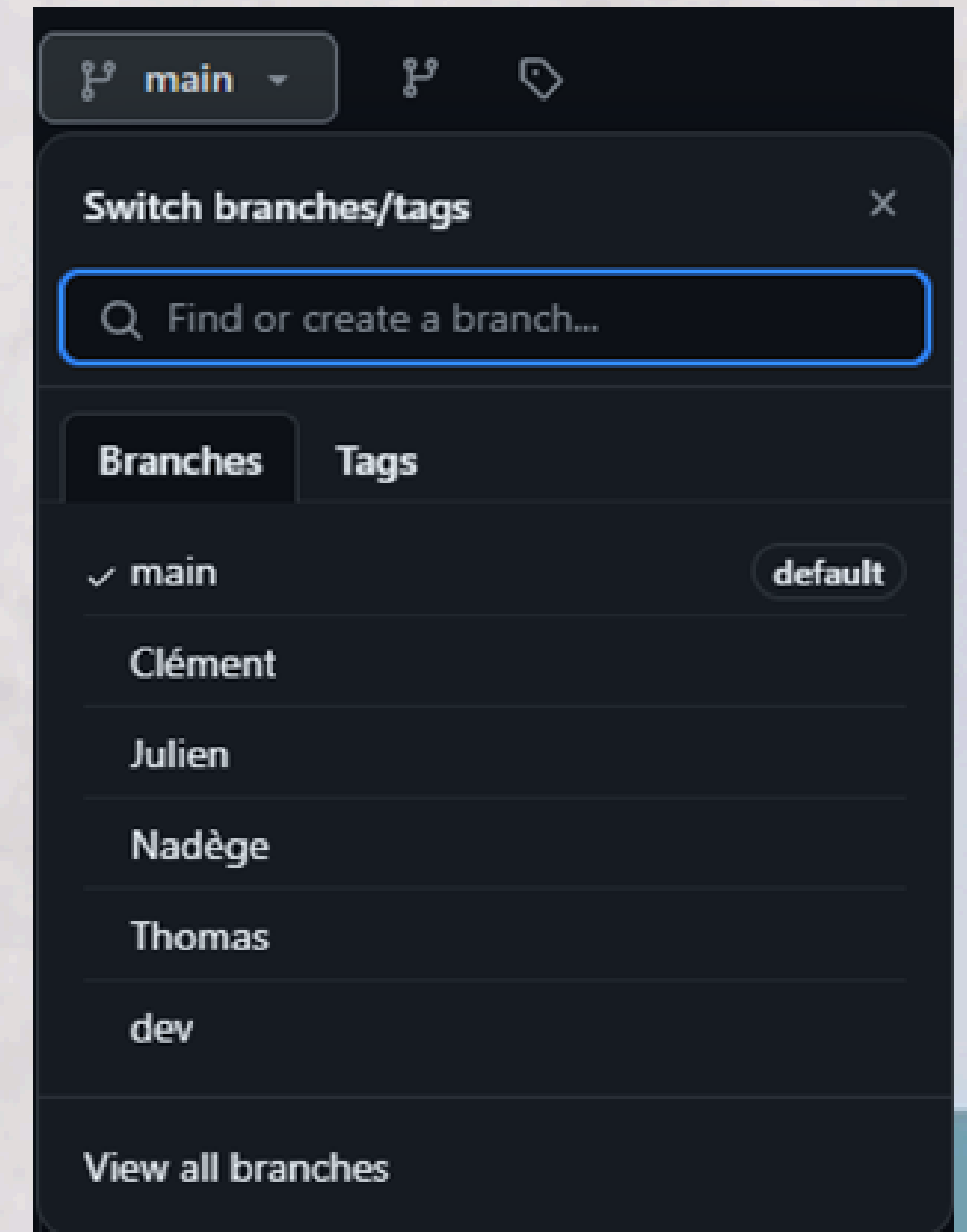
Gestion de projet

Versionning



Planification

Suivi



flake8



6

KANBAN ...

● A faire 2

Export la base de donnée

App Mobile React Native

+ Nouveau

● En cours 2

Dossier professionnel

Nadege Zongo

Clément Chevrier

Julien Fondu

Thomas

Gestion du score

Clément Chevrier

+ Nouveau

● En test 1

Créer les vidéos Animaker

Clément Chevrier

Julien Fondu

Thomas

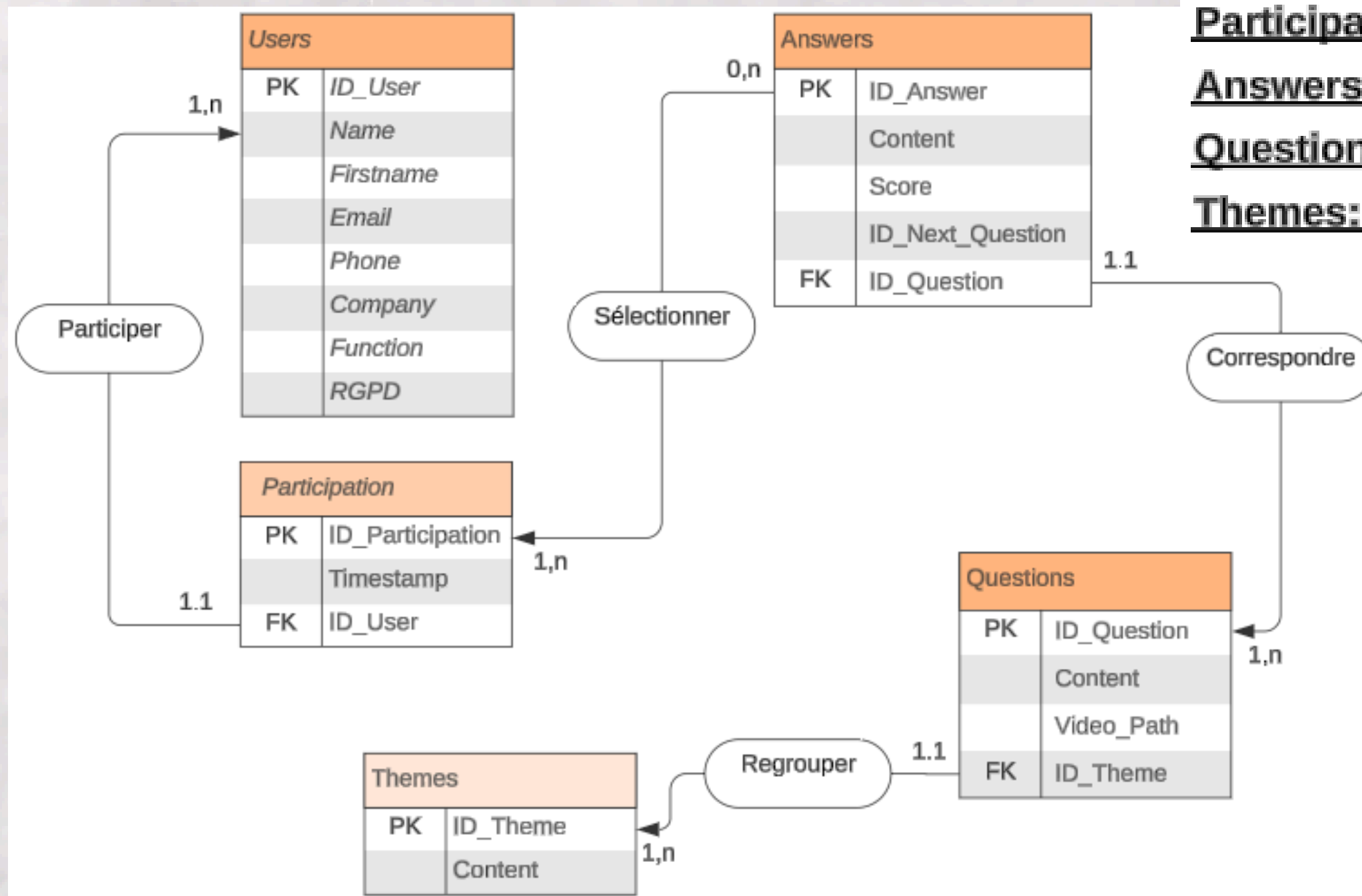
+ Nouveau

Grouper masqués

● Terminé 19

Réalisation

Conception de la base de données



Users: *ID_User*, Name, Firstname, Email, Phone, Company, Function, RGPD

Participation: *ID_Participation*, Timestamp, *ID_User*

Participation Answers: *ID_Answer*, *ID_Participation*

Answers: *ID_Answer*, Content, Score, ID_Next_Question, *ID_Question*

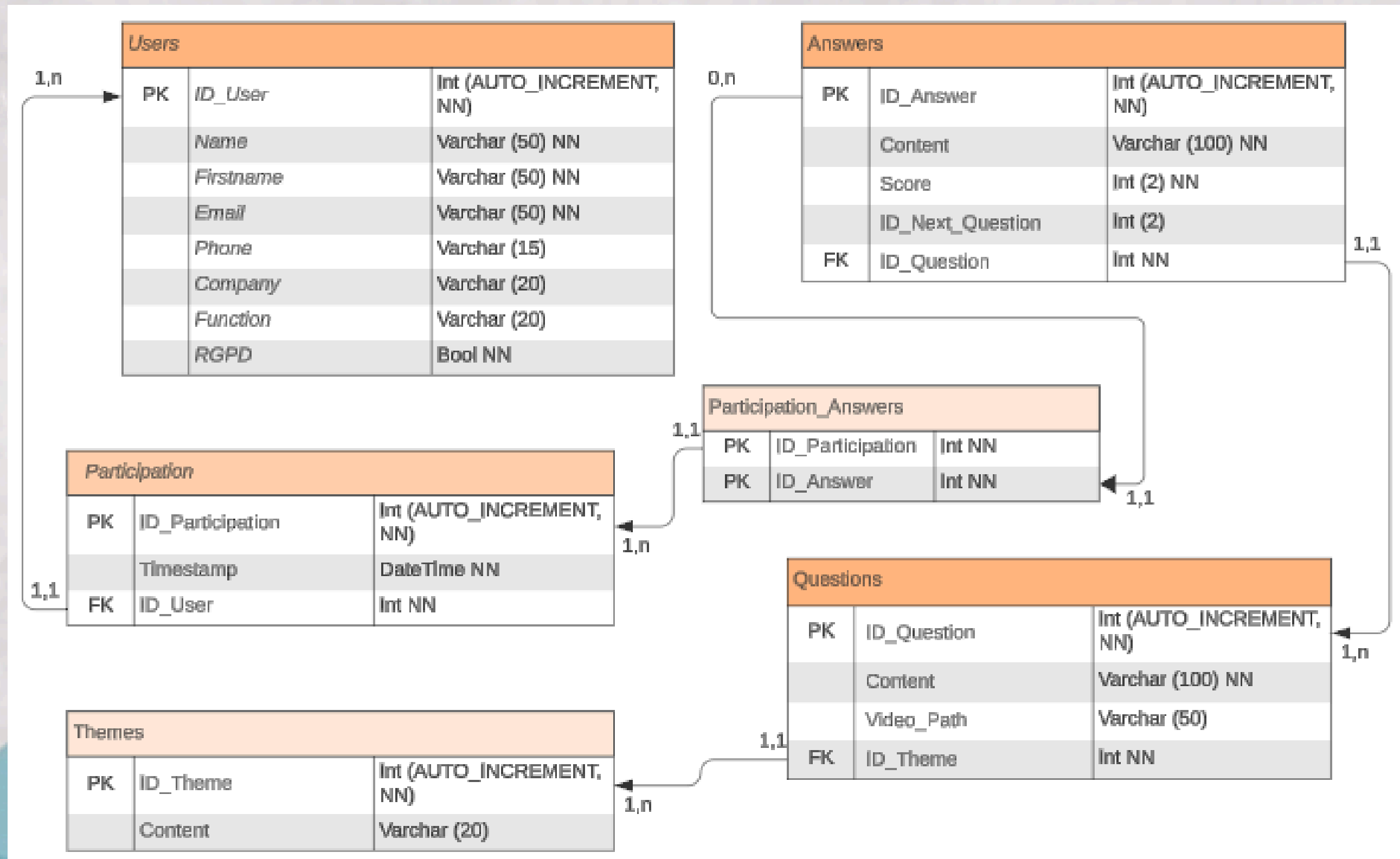
Questions: *ID_Question*, Content, Video_Path, *ID_Theme*

Themes: *ID_Theme*, Content

Merise (MCD, MLD, MPD)

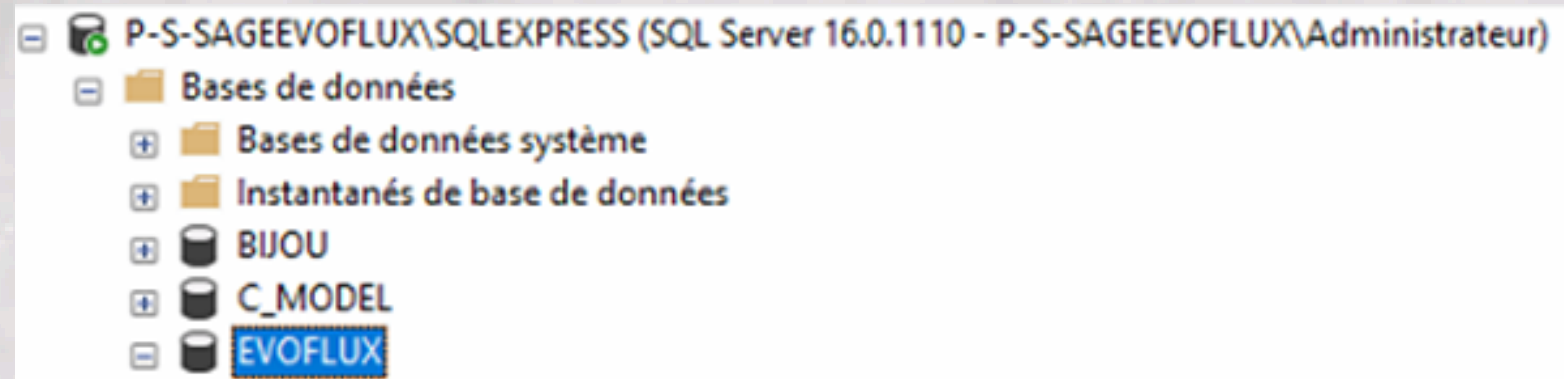
Réalisation

Conception de la base de données



Réalisation

Mise en place de la base de données



```
CREATE TABLE Users (  
    ID_User INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,  
    Name VARCHAR(50) NOT NULL,  
    Firstname VARCHAR(50) NOT NULL,  
    Email VARCHAR(100) NOT NULL,  
    Phone VARCHAR(15),  
    Company VARCHAR(20),  
    Function VARCHAR(20),  
    Conditions BOOL NOT NULL DEFAULT 0,  
    RGPD BOOL NOT NULL DEFAULT 0  
);  
  
-- Create Themes table  
CREATE TABLE Themes (  
    ID_Theme INTEGER PRIMARY KEY AUTOINCREMENT NOT NULL,  
    Content VARCHAR(20) NOT NULL  
);
```

DMZ

192.168.100.0/30



Pare-feu

Réseau local

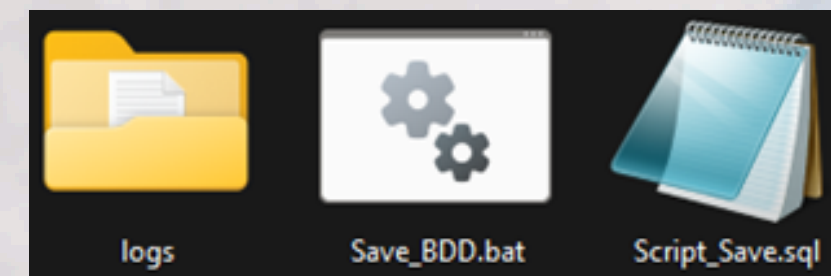
10.10.10.0/24



Internet

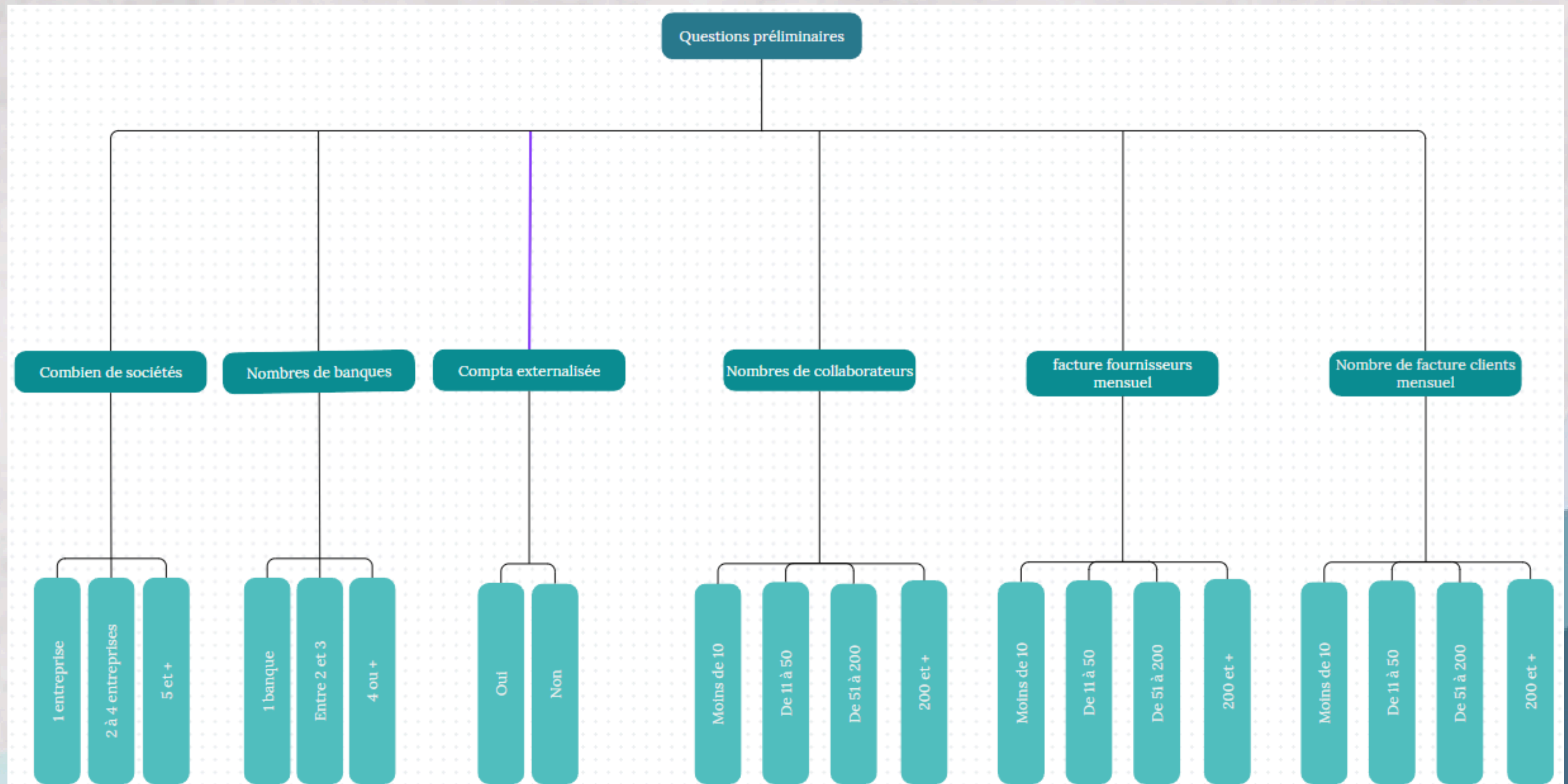


Backup automatique



Réalisation

Scénarios





Hugging Face

Lexica

Adobe
Firefly

runway

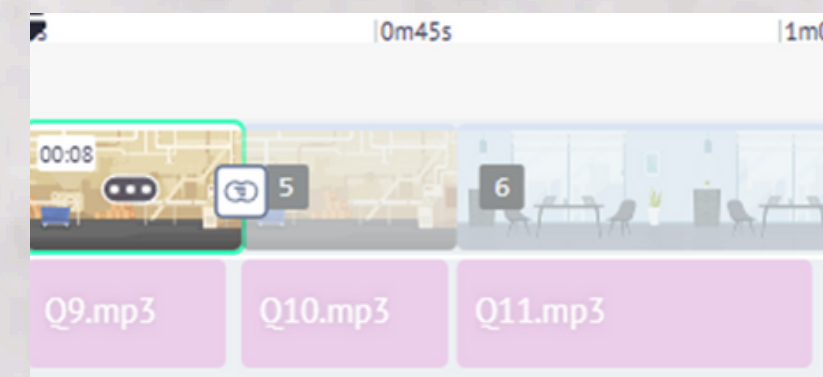
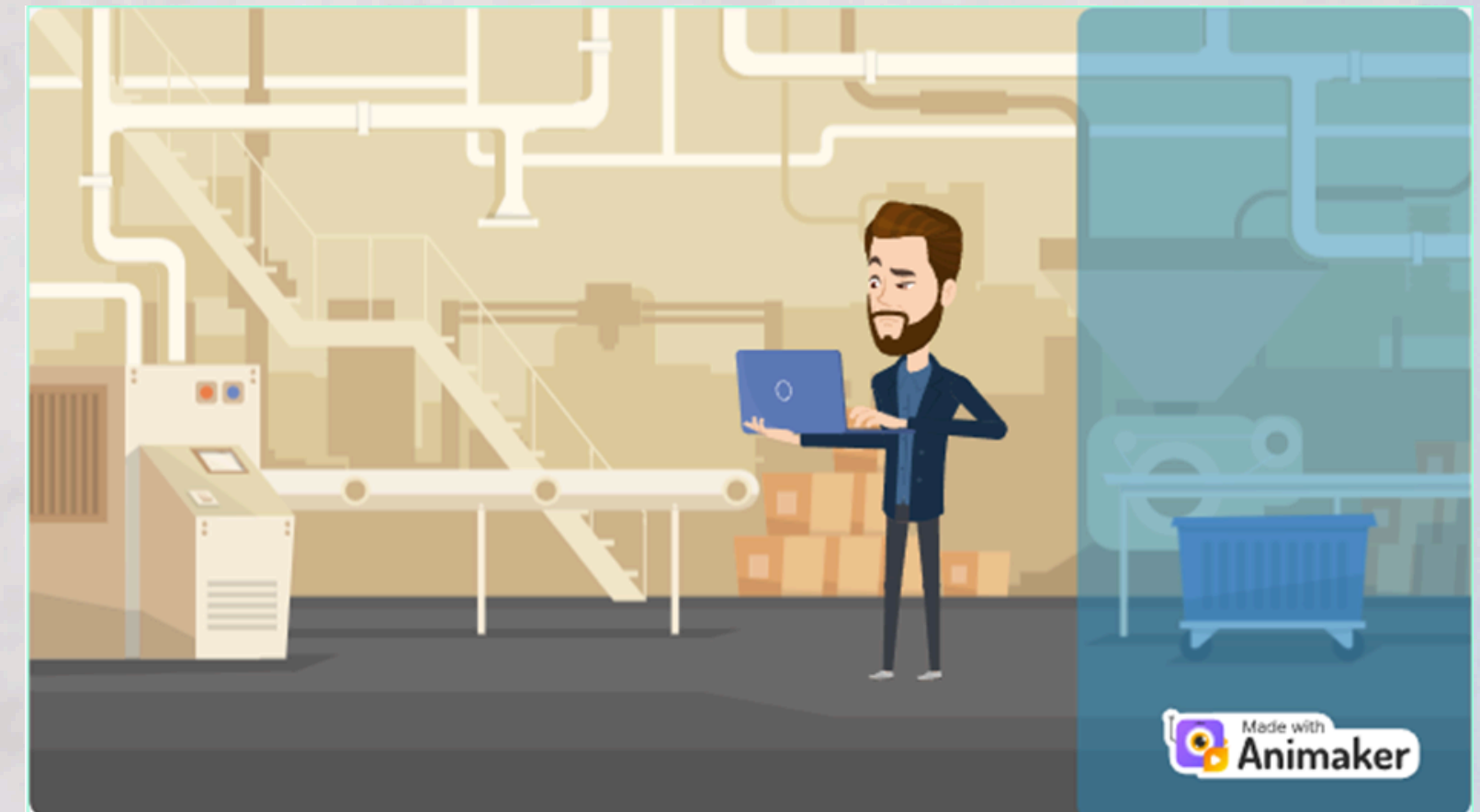
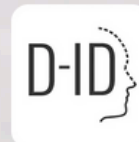
Visuels 1

Réalisation

Visuels 2



ElevenLabs



Réalisation

Accès aux données

CRUD

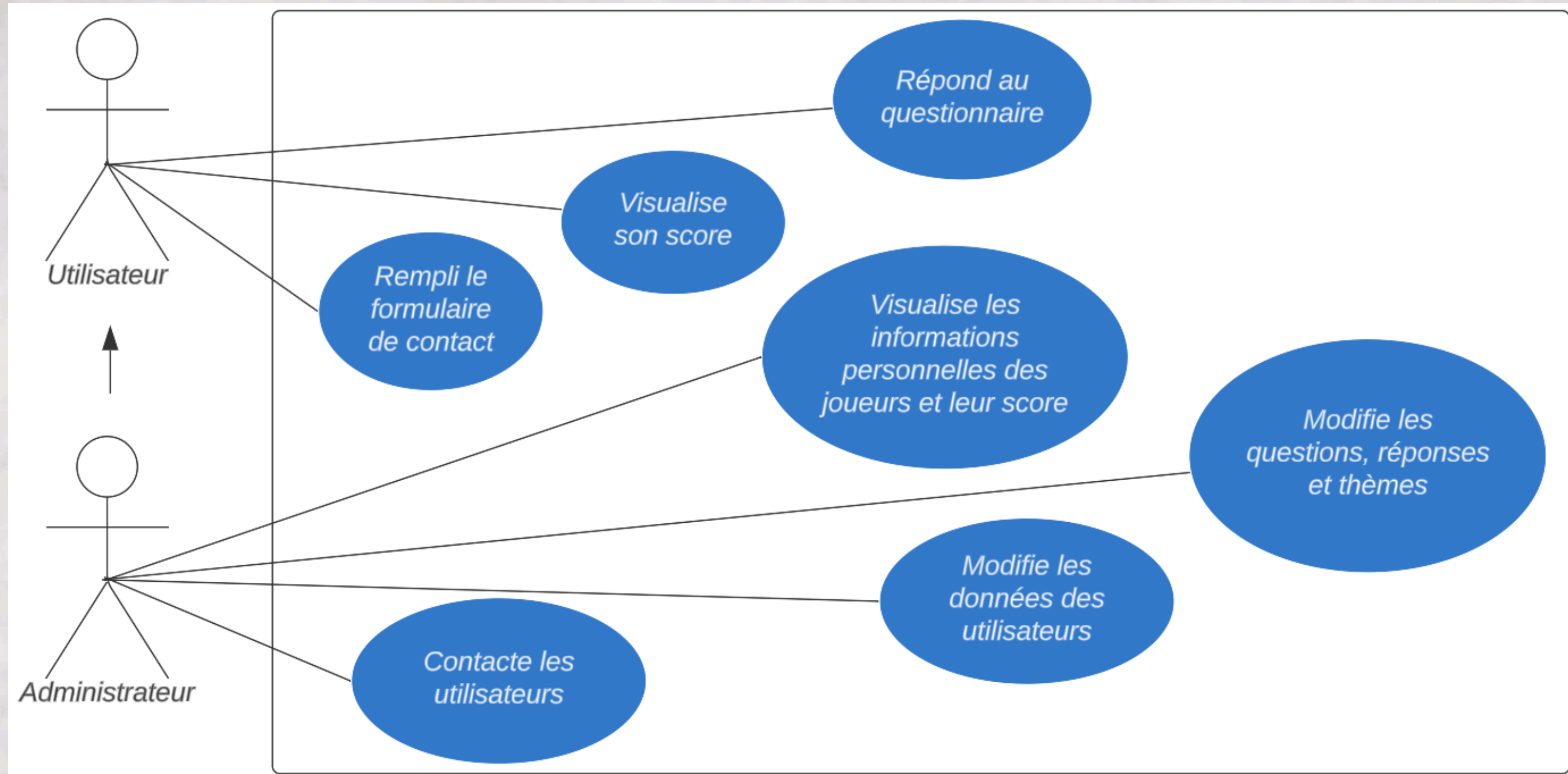
```
# Record the participant's choice, update Participation_Answers, and retrieve the ID of the next question
try:
    self.cursor.execute("INSERT INTO Participation_Answers (ID_Participation, ID_Theme, ID_Answer) VALUES (?, ?, ?)",
                        (participation_id, theme_id, choice))
    self.conn.commit()

    self.cursor.execute("SELECT ID_Next_Question FROM Answers WHERE ID_Answer=?", (choice,))
    next_question_id = self.cursor.fetchone()[0]
```

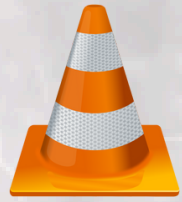
```
# Update user information in the Users table
self.cursor.execute("""UPDATE Users SET Name = ?, Firstname = ?, Email = ?, Phone = ?,
                        Company = ?, Function = ?, Conditions = ?, RGPD = ? WHERE ID_User = ?""",
                    (ecr_name, ecr_firstname, ecr_email, ecr_phone, ecr_company, ecr_function, conditions, rgpd, user_id))
self.conn.commit()
```

Fonctionnalités

Diagramme de cas d'utilisation UML



Bibliothèques

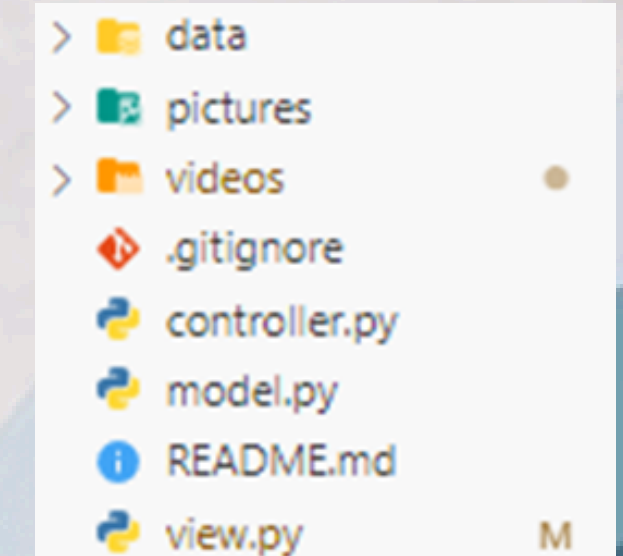
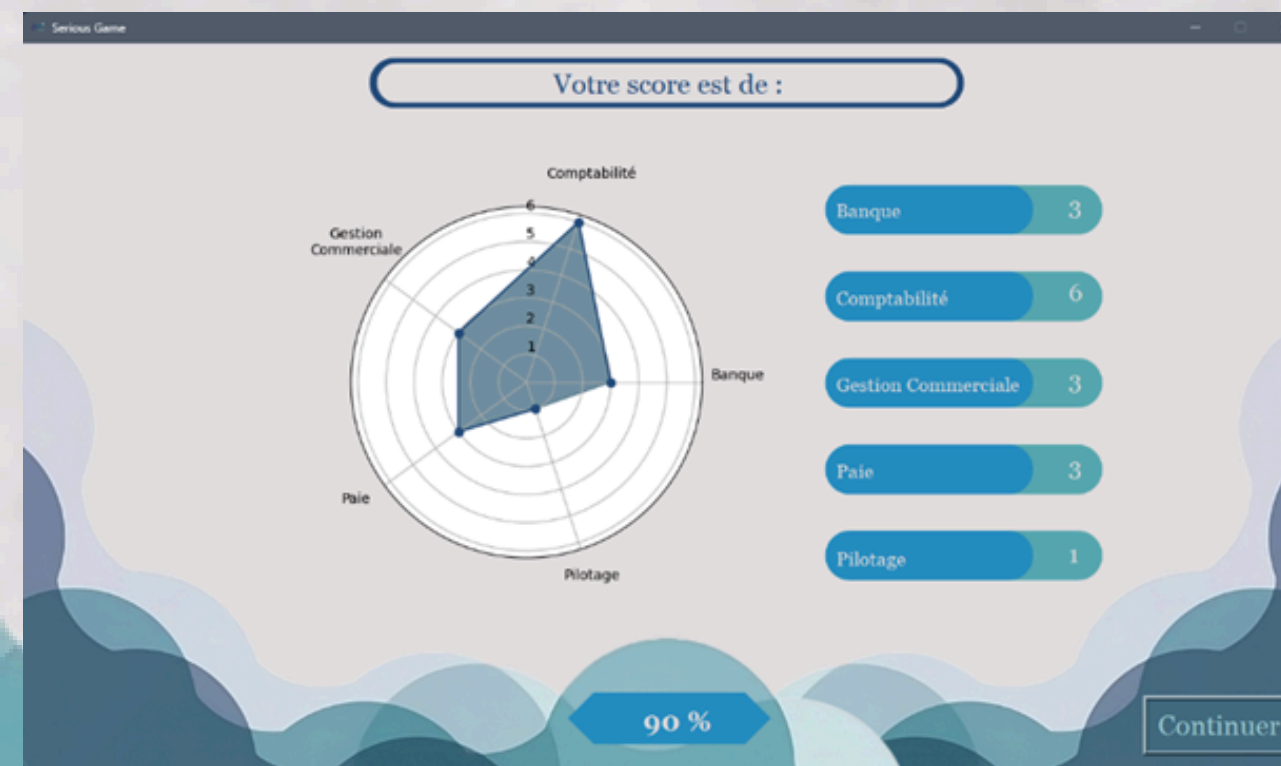
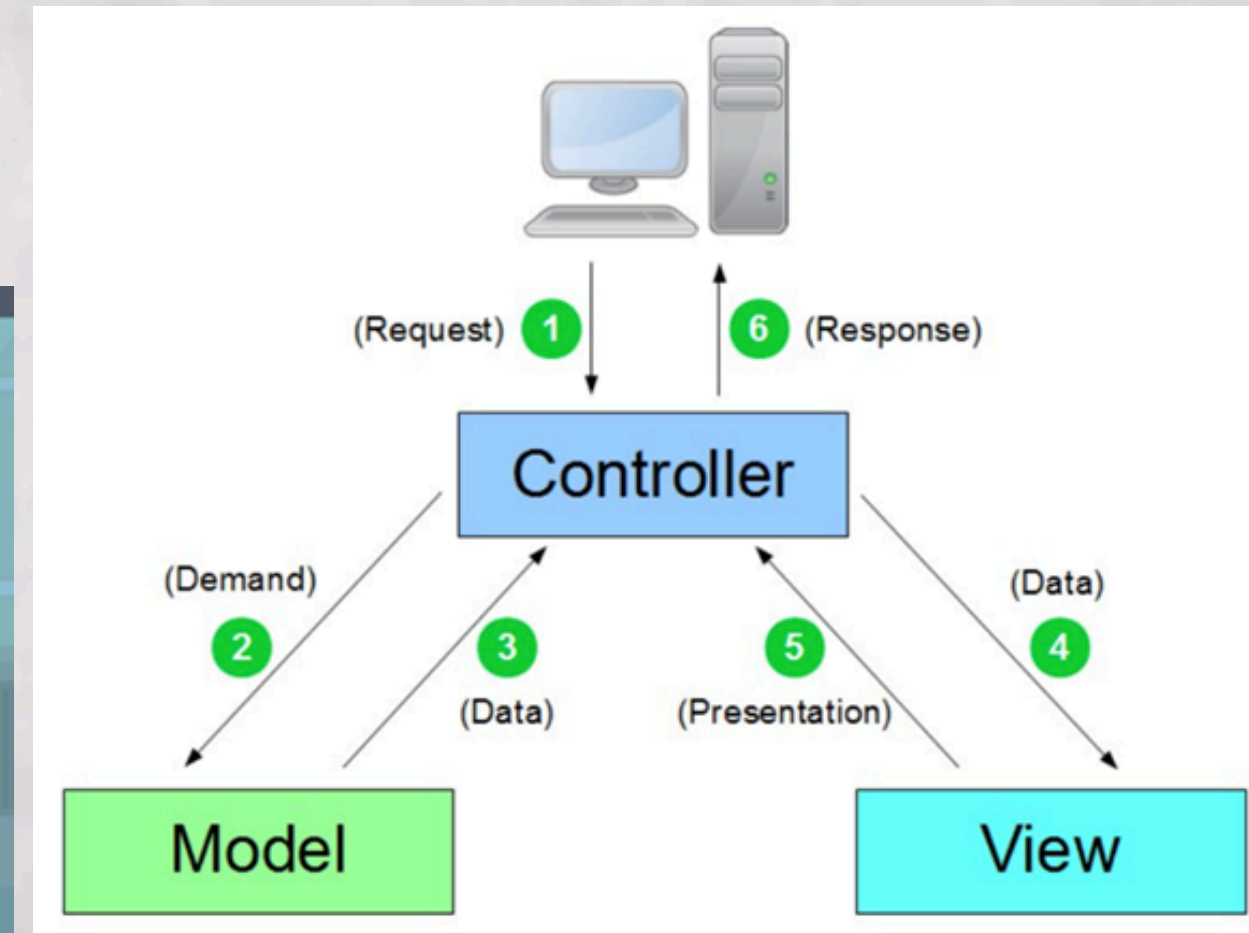


tkinter



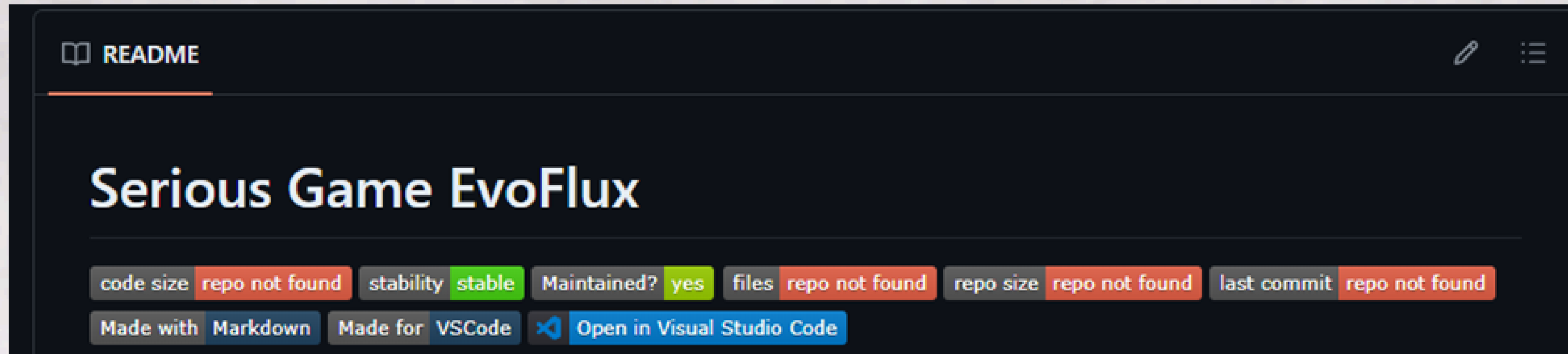
Réalisation Application Desktop

MVC



Réalisation

Installation et documentation



```
145 # Function to show a page with questions in the Tkinter window
146 def show_page(self, image_number, participation_id):
147     global choice_buttons
148     global button_exit
149
150     # Stop the previous video
151     if View.vlc_player is not None:
152         View.vlc_player.stop()
153
154     # Fetch Datas
155     current_dir = os.path.dirname(os.path.realpath(__file__))
156     question_text, choices, video_path = Model().fetch_question_data(image_number)
157     choice_buttons, disabled_button = self.create_choice_buttons(choices, image_number-1, participation_id)
```

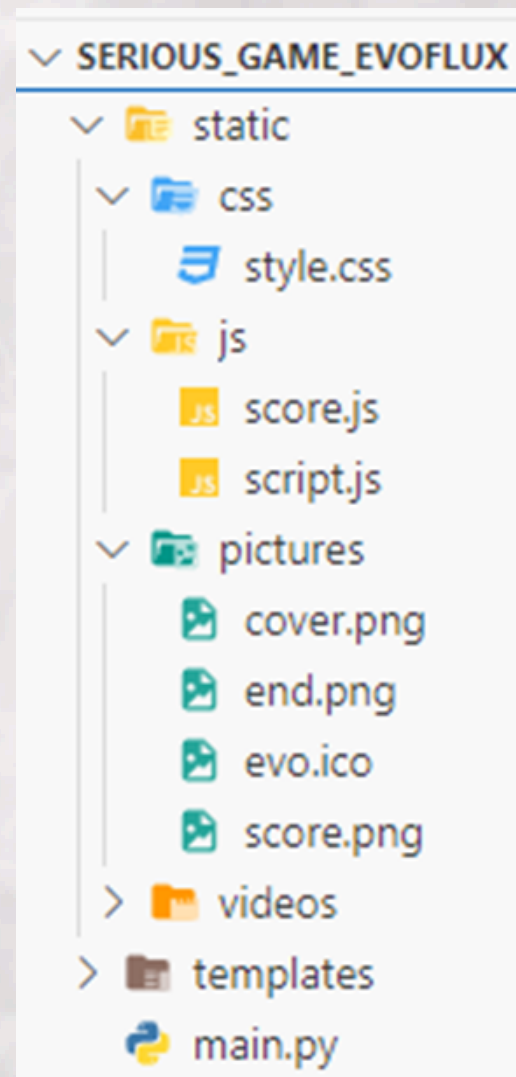
Reproduction de
l'application Desktop

Réalisation Site Web

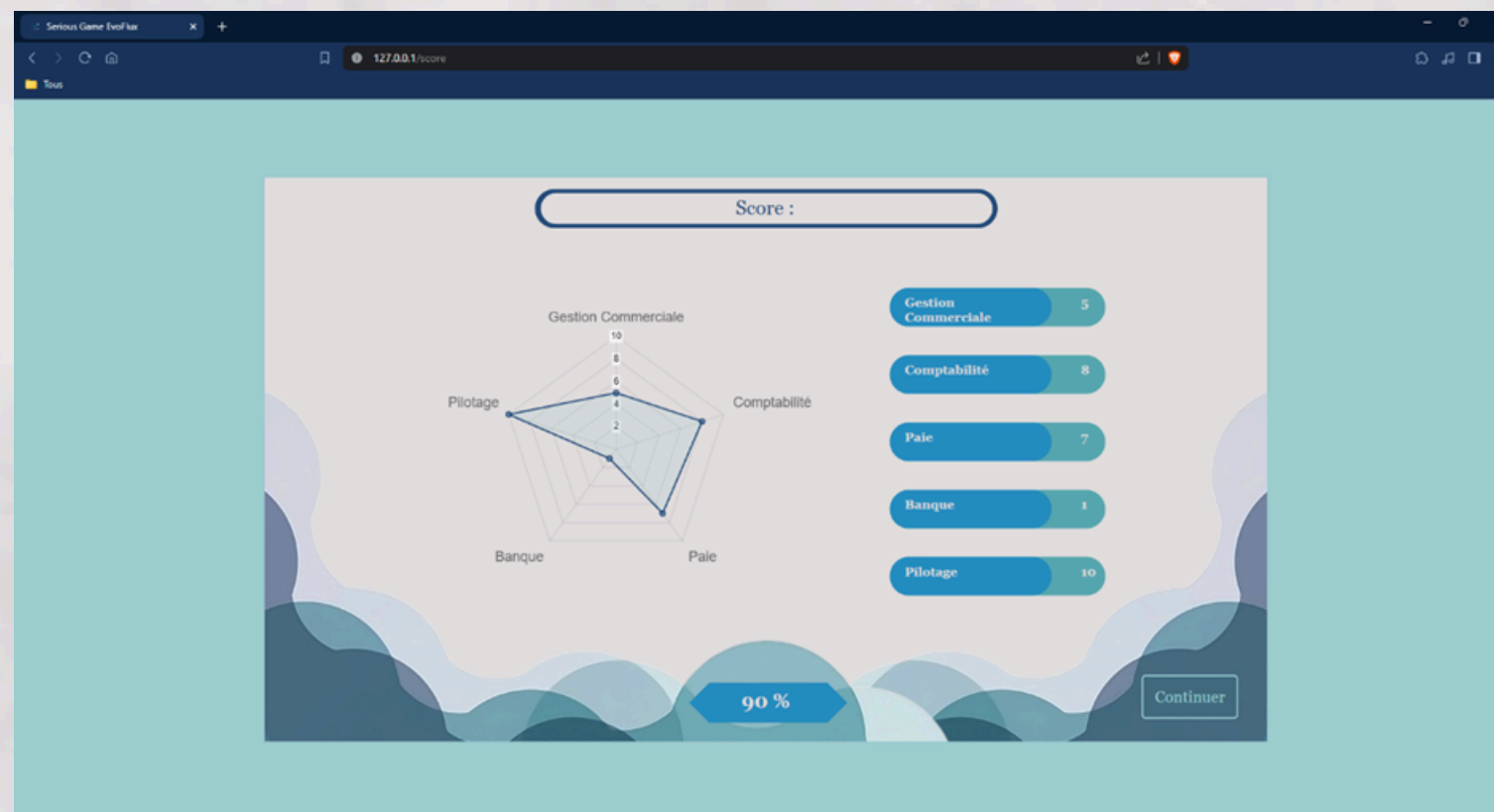
Front-End / Back-End

Flask

```
* Serving Flask app 'main'
* Debug mode: on
WARNING: This is a development server.
instead.
* Running on http://127.0.0.1:80
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 432-743-452
```



HTML / CSS / JS



```
main.py > Q12
You, hier | 1 author (You)
1  from flask import Flask, render_template
2
3  app = Flask(__name__)
4
5  @app.route('/')
6  def index():
7      return render_template('index.html')
8
9  @app.route('/Q1-6')
10 def Q1_6():
11     return render_template('Q1-6.html')
12
13 @app.route('/Q7')
14 def Q7():
15     return render_template('Q7.html')
```

Réalisation

Application Mobile

20:13 0.31 Ko/s



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Prénom * Clément

Email * clem.chevrier428@gmail.com

Téléphone 0683026777

Entreprise EvoFlux

Fonction Data Engineer

Message * Test

ENVOYER

```
PS C:\Users\Clément\Desktop\App EvoFlux\HelloWorld\backend> node server.js
Server running at http://localhost:3000
```

```
PS C:\Users\Clément\Desktop\App EvoFlux\HelloWorld> npx expo start
Starting project at C:\Users\Clément\Desktop\App EvoFlux\HelloWorld
```



> Metro waiting on exp://192.168.1.88:8081
> Scan the QR code above with Expo Go (Android) or the Camera app (iOS)

> Web is waiting on <http://localhost:8081>

> Using Expo Go
> Press s | switch to development build

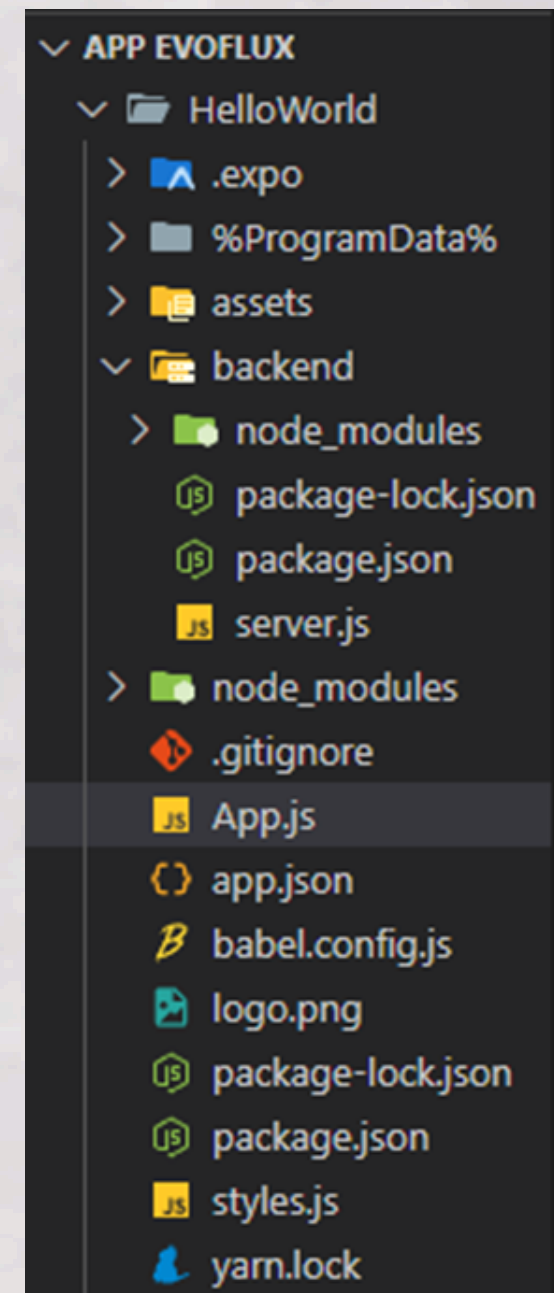
> Press a | open Android
> Press w | open web

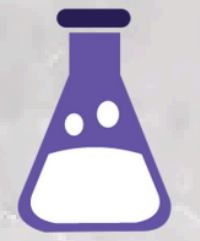
localhost:3000/submissions

Tous

Impression élégante ☒

```
{
  "submissions": [
    {
      "name": "chevrier",
      "firstName": "clément",
      "email": "clem.chevrier428@gmail.com",
      "phone": "0683026777",
      "company": "EvoFlux",
      "position": "Data Engineer",
      "message": "Test"
    }
  ]
}
```





unittest

Tests unitaires

```
# Test: Create a new participation
def test_generate_unique_participation_id(self):
    user_id = self.model.generate_unique_user_id('Doe', 'Jane', 'jane@example.com', '987654321', 'CompanyY', 'Manager')
    participation_id = self.model.generate_unique_participation_id(user_id)
    self.assertIsNotNone(participation_id)
    self.model.cursor.execute("SELECT * FROM Participation WHERE ID_Participation = ?", (participation_id,))
    participation = self.model.cursor.fetchone()
    self.assertIsNotNone(participation)
    self.assertEqual(participation[1], user_id)
```

```
PS C:\Projets\Serious_Game_EvoFlux> python -m unittest test_model.py
..
-----
Ran 2 tests in 0.003s

OK
```

Injectons SQL

```
# Check the length entry
length = len(text)
if int(min_length) <= length <= int(max_length):
    if is_email:
        # Check if the email is in a valid format using a regex
        email_regex = r'^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$'
        if re.match(email_regex, text):
            return True
```

Réalisation Tests et Sécurité



CNIL.

RGPD et Droits de réutilisation des données

```
if rgpd:
    # Update User Info into the Database if rgpd is check
    Model().update_user_info(user_id, ecr_name, ecr_firstname, ecr_email,
    | | | | | | | | ecr_phone, ecr_function, ecr_company, rgpd)
    print("Les informations de l'utilisateur ont été mises à jour avec succès.")
```

☐ J'accepte l'utilisation de mes informations
uniquement dans le but d'analyses ou afin d'être
recontacté.

Réalisation

Déploiement



Google Cloud Platform

RÉSUMÉ

MISE EN RÉSEAU

SÉCURITÉ

TESTS DE CONNECTIVITÉ

Mise en réseau

Nom de la connexion

vaulted-acolyte-381511:europa-west9:sql_connection

Connectivité IP privée

34.155.165.111

Connectivité IP publique

Désactivée

Adresse IP publique

Désactivée

Sécurité

Réseaux autorisés

1 network

Explorer

+ ADD

Type to search

Viewing workspace resources.

SHOW STARRED ONLY

vaulted-acolyte-381511

External connections

eu.bq_sql_connection

eu.bq_sql_connection

Connection info

Connection ID

projects/vaulted-acolyte-381511/locations/eu/connections/bq_sql_connection

Friendly name

Created

Mar 23, 2023, 4:08:07 PM UTC+1

Last modified

Mar 23, 2023, 4:08:07 PM UTC+1

Data location

eu

Description

Connection type

Cloud SQL - PostgreSQL

Cloud SQL connection name

vaulted-acolyte-381511:europa-west9:sql_connection

Database name

guestbook

Has credential

Yes

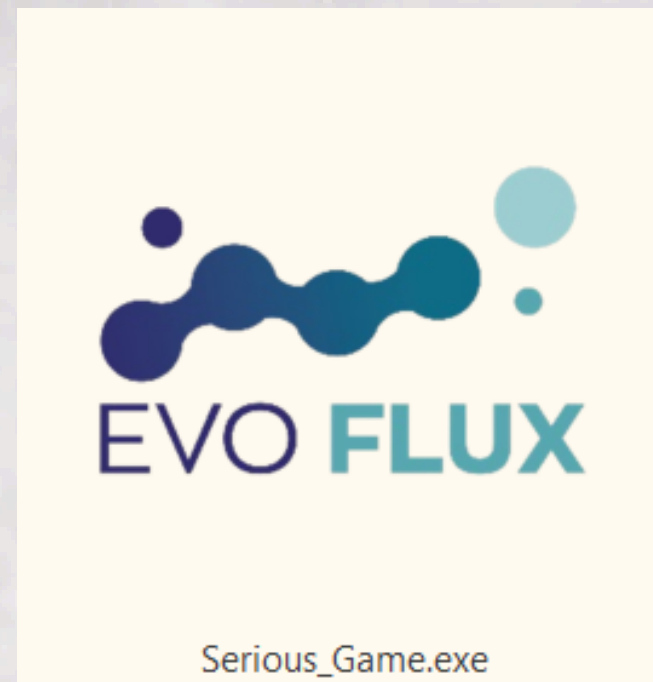
Service account id

service-711866656374@gcp-sa-bigqueryconnection.iam.gserviceaccount.com

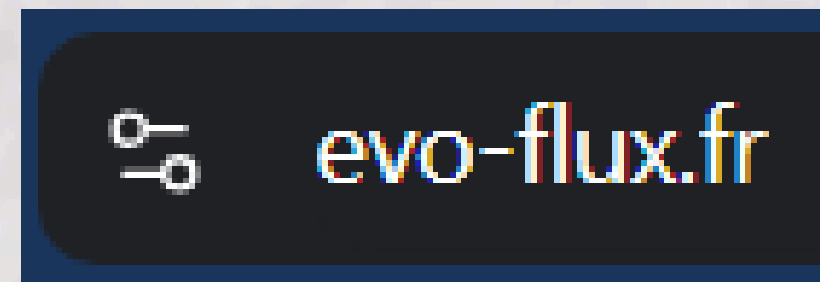
Réalisation

Déploiement

App Desktop



Interface Web



App Mobile



WORDPRESS



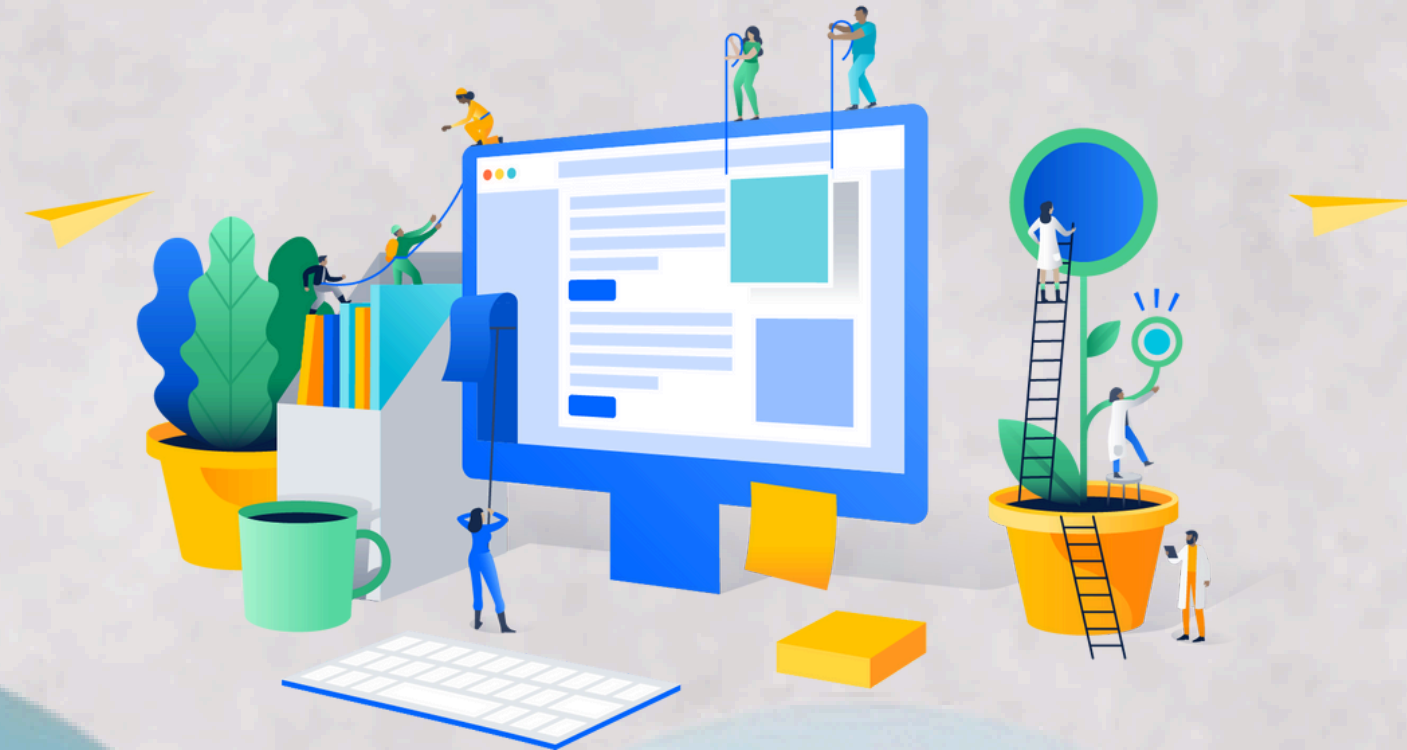
Conclusion

Finalisation + Incorporation au site web existant

1 difficulté principale rencontrée

Communication avec le client

Communication dans le groupe



MERCI

POUR VOTRE ATTENTION